Unit 1: The Marathon Runner: Review Sheet

Normal ranges

Oxygen saturation	95–100%
Blood glucose	90–130 mg/dL
Body temperature	37 °C
Plasma sodium	135–145 mEq/L (below = hyponatremia, the runner's problem)

Osmosis

Hypotonic outside	water moves IN, cell swells
Isotonic	no net change
Hypertonic outside	water moves OUT, cell shrinks

Cellular respiration

glucose + O_2 → CO_2 + H_2O + ATP (+ heat). Aerobic: uses O_2, mitochondria, lots of ATP. Anaerobic: no O_2, cytoplasm, little ATP, lactic acid (muscles) or alcohol + CO_2 (yeast).

Interactions Between Body Systems

Urinary system	Kidneys, bladder and urethra. Filters the blood, removes wastes like urea, and controls water and salt balance by making urine.
Excretory system	Removes metabolic wastes from the body: kidneys (urine), lungs (CO_2) and skin (sweat).
Circulatory system	Heart, blood and blood vessels. Carries O_2, glucose and hormones to cells and carries CO_2 and wastes away.
Nervous system	Brain, spinal cord and nerves. Detects changes and sends fast signals. The hypothalamus controls body temperature and thirst/ADH.
Muscular system	Muscles that move the body. Muscle cells use lots of ATP from cellular respiration and make heat (e.g. shivering).
Endocrine system	Glands that release hormones into the blood, e.g. the pancreas (insulin, glucagon) and ADH. Slower but longer-lasting signals.
Respiratory system	Lungs and airways. Brings O_2 into the blood (alveoli) and removes CO_2.

Homeostasis & Feedback Loops

Homeostasis	Keeping a stable internal environment (temperature, glucose, water/salt, O_2) even when outside conditions change.
Negative feedback loop	A response that reverses a change to bring the body back to its set point. Examples: insulin lowering blood glucose, sweating to cool down.
Positive feedback loop	A response that amplifies a change until an endpoint. Examples: oxytocin during childbirth, platelets during blood clotting.
Thermoregulation	Keeping body temperature near 37 °C. Hot: sweating and vasodilation. Cold: shivering and vasoconstriction. Controlled by the hypothalamus.

Osmosis

Cell membrane	The thin, selectively permeable boundary around a cell that controls what goes in and out.
Osmosis	Diffusion of water across a membrane from where there is more water (less solute) to where there is less water (more solute).
Hypertonic solution	More solute (salt) outside the cell than inside. Water leaves the cell and the cell shrinks.

Hypotonic solution	Less solute (salt) outside the cell than inside. Water enters the cell and the cell swells and may burst.
Isotonic solution	Equal solute inside and outside. Water moves in and out equally, so the cell stays the same size.

Energy & Cellular Respiration

Glucose	A sugar ($C_6H_{12}O_6$) whose chemical bonds store chemical potential energy. The main fuel for cellular respiration.
Mitochondria	The organelle where aerobic cellular respiration happens and most ATP is made.
ATP	The molecule cells use directly for energy. Made by cellular respiration.
Cellular respiration	Cells breaking down glucose to release its energy as ATP: glucose + O_2 → CO_2 + H_2O + ATP (+ heat).
Aerobic respiration	Cellular respiration that uses oxygen. Happens in the mitochondria and makes lots of ATP.
Anaerobic respiration	Cellular respiration without oxygen. Happens in the cytoplasm and makes much less ATP. Produces lactic acid in muscles, or alcohol + CO_2 in yeast.

Also from class

Insulin	Hormone from the pancreas that lowers blood glucose: cells take in glucose and extra is stored as glycogen.
Glucagon	Hormone from the pancreas that raises blood glucose by breaking glycogen back down into glucose.
Glycogen	The storage form of glucose, kept in the liver and muscles.
Antidiuretic hormone (ADH)	Hormone that tells the kidneys to keep more water. More ADH means less, more concentrated urine.
Osmoregulation	Keeping the right balance of water and solutes so cells stay isotonic with body fluids. Done mainly by the kidneys.
Hyponatremia	Blood sodium too low (below 135 mEq/L), often from drinking too much plain water. Cells swell, which in the brain causes confusion and coma.
Vasodilation / vasoconstriction	Blood vessels near the skin widening (release heat) / narrowing (keep heat in).
Chemical potential energy	Energy stored in chemical bonds (like glucose), released during a chemical reaction.